

Enhancing Students' Social and Emotional Learning in Educational Virtual Heritage through Projective Augmented Reality

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Figure 1: A) A functional WaS prototype used in the shelter Refugi 307; (B) Student during the Pictionary activity, (C) A collage illustrating students' understanding of the Spanish Civil War.

ABSTRACT

The global agenda in education foresees the integration of social and emotional learning to equip students to succeed in our evolving digital society. In this paper, we focus on deepening into the research on AR/MR technologies to foster students' comprehension of socio-cultural values in historical contexts. We address this challenge by exploring the potential of the interaction paradigm called World-as-Support (WaS). We present a case study of an educational Virtual Heritage experience with primary students for a bomb shelter built during the Spanish Civil War. Our findings showed that the experience enhanced students' capabilities to reflect upon high-level issues related to value human dignity, to grasp some of the essential qualities of the value solidarity and to connect historical events with present political situations in Spain. Finally, we discuss four design recommendations for learning activities based on the WaS concerning (1) the enhancement of students' competences in collaboration, and communication; (2) critical thinking; (3) contextualization of historical contents; and (4) moral and ethical considerations for digital augmentation.

CCS CONCEPTS

• **Human-centered computing** → **Mixed / augmented reality.**

KEYWORDS

Mixed Reality, Augmented Reality, Embodied Interaction, Socio-Cultural Values, World-as-Support, Children, Social and Emotional Learning, Virtual Heritage, Moral and Ethical Considerations

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1 INTRODUCTION

In 2016 the World Economic Forum published a report that focused on students' skills gap in the 21st century and ways how technology may be invaluable at enabling social and emotional learning (SEL) [2]. According to this report, students lack crucial social and emotional competencies such as communication, collaboration and critical-thinking. These skills are critical for future generations to prepare them for the demands of a rapidly changing work environment, position them to achieve better academic outcomes and support them to contribute to society. Teaching these competencies are no longer only the matter of employing technology-enhanced learning strategies in classrooms. Curators and educators of history museums and archaeological sites increasingly consider their

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mission to provide educational activities that focus on cultural and civic literacy to transmit fundamental socio-cultural values. This literacy implies students' development of knowledge and skills to understand the traditions and history of a group of people from a given culture [13]. At the same time, it includes the knowledge to participate effectively in civic life through understanding governmental processes, and knowing how to exercise the rights and obligations of citizenship at local, national, and global levels. In other words, students also acquire an understanding of the local and global implications of civic decisions [1]. In this request, the potential of Augmented and Mixed Reality (AR/MR) mobile technologies has attracted researchers in both academia and industry [21, 23, 29]. Research in this field highlights the benefits of embodied interaction and collaborative learning experiences in enhancing SEL with respect to other interaction modalities [23]. However, these studies also stressed that there is still a need for a deeper understanding of AR/MR specificities to design systems capable of unfolding students' skills in social and cultural awareness [27]. According to the report of the World Economic Forum, the acquisition of these skills entail the understanding of socio-cultural values by obtaining greater empathy, respect and tolerance for others and the comprehension of the value of cultural self-identity [2]. This paper aims at deepening into the research on AR/MR technologies to foster students' cultural and civic literacy and comprehension of socio-cultural values in historical contexts. Specifically, we focused on investigating the potential of an emerging interaction paradigm, called World-as-Support (WaS) [16], which proposes the use of protective technologies to visualize computer-generated images directly on the physical environment. Our goal in this case study is to answer the following question: *How can we enhance students' SEL in outdoor cultural heritage sites using the WaS interaction paradigm?*

The contributions of our work are as follows: (1) New insights about how the WaS interaction paradigm can be used in the context of educational Virtual Heritage (VH); (2) A multi-method approach for analysing children's learning about SEL and historical contents; (3) example learning activities based on WaS illustrated in a case study of a VH learning experience. Our findings are based on a study with 16 primary students of a public school in Barcelona. We show how educational strategies enhanced students' SEL of the learning context. To conclude, we discuss four design recommendations for educational technologies based on the WaS interaction paradigm to complement and extend traditional learning experiences in the context of cultural heritage concerning **(1) the enhancement of students' competences in collaboration, and communication; (2) critical thinking; (3) contextualization of historical contents; and (4) moral and ethical considerations for digital augmentation.**

2 BENEFITS OF AR/MR TECHNOLOGIES IN CULTURAL HERITAGE EDUCATION

AR/MR technologies have shown to play a pivotal role in fostering SEL [2] to complement and extend traditional learning experiences. These innovative approaches are capable of mixing the physical and virtual worlds and facilitating forms of human interaction that were still impossible a decade ago. During last years,

several learning experiences have been developed for museums and cultural heritage sites addressing different aspects of learning such as fostering students' skills towards the understanding of past events [23, 32], culture and identity [30], perspective-taking [12, 20], critical thinking [22], and collaboration [5, 14], among others. To achieve these educational goals, scholars have investigated the benefits of a wide range of interaction modalities with young visitors. For instance, Sakr et al. [23] investigated how emotional engagement had an impact on children's learning experience about events related to World War II. Students were engaged with a set of location-based tasks while using a digital environment designed for the iPad. Stanton et al. [31] researched the benefits of adult-child interaction to facilitate children's discovery, reasoning and reflection upon historical places and events. The VH experience *We Hunters* [12] invited children to experience and learn about hunting strategies which were illustrated in a cave painting by using Full-Body actions. This helped them to "embody" certain enactments of people from ancient cultures and to better contextualize underlying socio-cultural meanings in relation to the museum exhibition. Halloran et al. [11] showed how digital augmentations during educational field trips can support creative writing about historical contexts. Pittarello and Bertani [19] presented an approach for using the environmental and social context for augmenting the emotional engagement of story listeners. Bouvin et al. [4] used a geospatial hypermedia system to support project based education and learning outside of the classroom through contextualization of information. Häkkinen et al. [10] highlighted the importance for offering users their own private view of virtual graveyards instead favouring collaborative experiences. Designs for VH experiences in these context need to address with special care and consideration due the sensitivity of the context. In this regard, Correll [8] pointed towards the moral components of the design and use of visualizations. Specifically, the author stressed designers' and researchers' responsibilities of carefully choosing visual contents and the impacts it may have on the users.

3 CASE STUDY

Our study was carried out in the context of a cultural heritage location; namely *Refugi 307*. The site is one of the 1,402 bomb shelters that were built by civilians during the Spanish Civil War in Barcelona aimed at protecting the population (Figure 1). Today, the shelter is part of the Barcelona History Museum which provides guided visits through the site. The main goal of the project was to explore new possibilities to complement the current guided visit with an educational VH experience. One purpose of the present study was to deepen research on how a VH learning experience based on the WaS shaped students' cultural and civic literacy and understanding of socio-cultural values related to the Spanish Civil War. We decided to focus on cultural and civic literacy because during the initial definition of the project requirements, the curators and educational experts had stressed the importance of teaching socio-cultural values in history education because this knowledge could prevent that students repeated mistakes from the past [24, 28]. Furthermore, the national curriculum of Spanish primary schools foresees teaching socio-cultural values as an essential mission to appreciate norms of our living together, learn to act in accordance with

them, prepare for the active exercise of citizenship by respecting and defending human rights, as well as the pluralism of a democratic society [7]. In this study, we understand the concept of values as the fundamental elements of culture. They define the meaning and significance of people within a society. The rules and norms of a society are derived from its values [9]. Building on this definition, we focused on three socio-cultural values that were relevant to the understanding of the educational and historical context as follows:

- **Human dignity** is understood as the rights of a person to be valued and to be ethically tried;
- **Solidarity** is understood as the agreement between and support for the members of a group bonded together due to their same interests, values, and sympathies;
- **Pluralism** is understood as the existence of different types of people who have different beliefs and opinions within the same society.

In the following section, we describe the design features of the prototype, the procedure and data analysis of the study.

3.1 Design features of the prototype

For the study, we used a functional prototype based on a marker-recognition system. We used the markers strategically to trigger contents in specific locations of the shelter in a simple manner. The device recognized markers in physical space and projected virtual content next to the marker on any type of surface. It was composed of an Android Aquaris BQ tablet, a Bluetooth loudspeaker, and a pico-projector. All components were enclosed in a specially designed casing for our prototype (Figure 2). We implemented six interactive learning activities in the WaS experience (Table 1). Five of the learning activities had been explored in previous studies [26, 27] using low-tech material and a mid-tech prototype of the WaS system. For these activities, our results had indicated a high potential in enhancing student's collaborative learning and perspective-taking in relation to emotional aspects of Spanish war civilians. We employed them again in this study to observe their potential using a functional prototype.

3.2 Procedure

In this study participated 16 primary students in total (girls = 7, boys = 9; mean age = 11.27 years). They were students from the same class of public school in Barcelona. Before the study, we provided the legal tutors of the students with a consent form in which they were informed about and asked for their agreement on data collection and dissemination. The learning contents and materials shown to the children were chosen and elaborated together with the curators and museum educators [24, 28]. Our study was organized in three consecutive sessions during the same week. The first session was held in the school and had a duration of 60 minutes. Two researchers and one teacher facilitated the activities during the session. Its main goal was to introduce the children to the learning context and to evaluate their previous knowledge and comprehension of socio-cultural values related to it. First, we handed out a questionnaire to each child aimed at assessing their understanding of the educational context. The questionnaires were based on open-ended questions that children were asked to complete such as "Which situations did the civilians experience during the war?

How did it make them feel?" The children were encouraged to give multiple answers. We then facilitated an activity based on the board game Pictionary. In previous studies, we had used this technique as a playful strategy to elicit and understand children's previous knowledge of conceptual learning goals [25]. For this study, the researchers selected a set of terms and concepts related to the addressed socio-cultural values. The terms were previously discussed with the teachers and museum educators to focus them on the main educational goals of the visit. The terms used for the concept *human dignity* were: war, refugee, collective trauma, soldier; for *solidarity*: individualism, citizen association; for *pluralism*: protest, coup, exile, passive defence, dictatorship. The second session was conducted during the visit in the shelter. We used two devices of the prototype that were randomly assigned to the students in the different interactive learning activities. Each child had the opportunity to use the WaS system during an activity at least one time. The guide explained the Spanish Civil War in a chronological order and highlighted related aspects and events through physical characteristics of the shelter or traces of missing objects. The same learning content was complemented with the performance of the six interactive activities (Table 1). The visit had a duration of 60 minutes. After the visit, we handed out a questionnaire to each child. The goal was to research how their worldviews around the educational context had changed. Therefore, we asked them "How do you feel about the civilians of the Spanish Civil War after the guided visit?" The third session took place in the school and had a duration of 90 minutes. First, the students were randomly divided into groups of four to five members. We gave each group a different term that was related to one of three socio-cultural values (human dignity – collective trauma; solidarity – citizen association; pluralism – passive defence). We provided each group with the same fifteen photos depicting different situations during the Spanish Civil War. A description of each photo is reported in (Appendix 2). The material was provided by the library of the Barcelona History Museum and selected together with the museum educators. The students were instructed in choosing pictures that would represent their understanding of the different terms and to complement them with a written description. The researchers then went around with cameras and recorded a short interview with each group about their reflections. The purpose of this activity was to evaluate if the children had a deeper understanding of the concepts related to socio-cultural values after visiting the shelter and participating in the learning activities using the WaS system.

3.3 Data Analysis

To evaluate how the WaS interaction paradigm shaped students' understanding of socio-cultural values and their experience with the functional prototype, we triangulated data from the analysis of the *open-ended questionnaires*; the analysis of *video recordings of their in-situ interactions* during the guided visit; the analysis of *drawings*; the analysis of the *collages* and the analysis of *interviews*. All data was first individually analysed by two researchers. They compared and discussed then their individual results until a common agreement on the findings was achieved. The procedure was as follows. The open-ended questionnaires were first transcribed by one researcher. Two researchers analysed then written contributions



Figure 2: Overview of the system design of the Was Interaction paradigm.

Table 1: Overview of the activities using the WaS prototype during the guided visit

Activity	Description
<i>A1: Bomb Attack</i>	One student pointed with the WaS system at a marker at the ceiling of shelter. A video of an airplane dropping a bomb was displayed.
<i>A2: Behavioural rule sign</i>	One child pointed on the marker at the wall and a picture of an old sign containing the rules of the shelter was displayed. The guide asked the students to read out aloud the content of the sign.
<i>A3: Radio documentary</i>	One child pointed on a marker in a shelf. The image of an old radio was displayed. The group listened then to an extract of a radio program that was broadcasted during the Spanish Civil War.
<i>A4: Infirmary</i>	One student pointed at a marker placed on the entrance of the infirmary and one part of an image representing a stretcher was displayed. Another student pointed then on this image and the other part of the stretcher was displayed. They were asked to synchronize their movements and to take a virtual person on the projected stretcher safely into the infirmary.
<i>A5: Construction of the shelter</i>	Eight students were divided into two groups. In each group, one child was in charge of the WaS system. The other three students received a model of a brick that was used as a prop for the activity. They were asked to place each brick in front of a different marker and to point with the WaS system on it. This way, step by step, a virtual arch of a new part of the shelter was displayed.
<i>A6: Children's drawings about the war</i>	Two children compared two different images. One child pointed on a marker at the wall and a child's drawing from the late 1930s was displayed. Another child pointed on a different marker and projected a child's drawing from Syria next to the first one.

using the NVivo software. We extrapolated results according to the three common values of the European Union defined as *human dignity*, *solidarity*, and *pluralism* [9]. We compared student's previous worldviews with those after the visit. Subsequently, we watched all videos of the sessions and analysed interviews of the drawings and interviews of the collage activity. The researchers visualized the selected videos in slow-motion several times, with and without audio, to transcribe the resources employed by the children in a descriptive format. We also made screenshots of significant body actions and organized them in a storyboard format. For the analysis, we focused on the following resources: body postures, facial expressions, gaze, verbal expressions, and social and spatial interactions. To analyse the drawings, we employed a mixed approach that combined content analysis and multimodal analysis according to the approach presented in Malinverni et al. [17, 18]. This approach focuses on the analysis of children's multiple embodied resources represented of the same aspects (e.g. through drawings, verbal and bodily expressions). To analyse the interviews, we employed a content analysis approach. Specifically, one researcher

completely transcribed the interviews. The transcripts of the interviews about the collages were analysed using the coding scheme based on the three common values: human dignity, solidarity, and pluralism. To analyse the collage, we looked at which photos the students used in each collage and the aspects they described in their written and verbal contributions. Finally, we merged (1) the data of in-situ interactions, drawings and interviews to interpret the emotional experience and interactions during the activities and (2) the data of open-ended questionnaires, Pictionary Technique, collage and interviews to grasp children's changes in socio-cultural values.

4 RESULTS

In this section, we report on the results obtained for the students during the WaS activities (Figure 3). We focused on children's emotional experience, interactions and conceptual changes of socio-cultural values of the educational context (human dignity, solidarity, and pluralism).

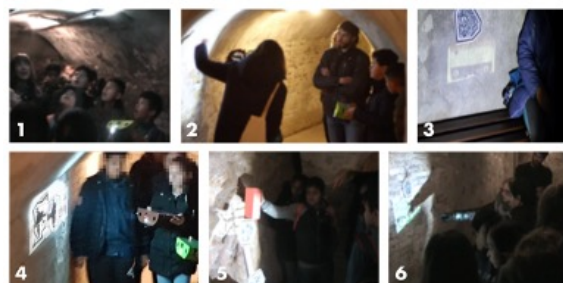


Figure 3: The six educational activities based on the WaS prototype: (1) bomb attack; (2) behavioural rule sign; (3) radio; (4) infirmary; (5) construction of the shelter; (6) children's drawings about the war.

Students' emotional experience and interactions during the WaS activities. Our findings showed that all activities complemented the guide's explanations. Second, activities in which one student led the tasks and others joined as active spectators (A2, A3) and role-play activities (A4, A5) supported group participation and group discussions. Third, the students' were particularly emotionally impacted by activities that either used audiovisual materials (A1) or implied role-play activities (A5, A6). Four, activities performed as a role-play in pairs or in group helped to support students abilities to step into other people's lived experiences in the past (A4, A5, A6) and understand effortful tasks that they had performed (A4, A5). Finally, the activities directed students' attention to physical characteristics of the shelter that were important for understanding the historical context (A1, A4, A5). More specifically, our findings based on the in-situ interactions and interviews depicted that particularly three activities had a strong emotional impact on the children and helped them to empathize with civilians from the Spanish Civil War. First, during the activity *bomb attack*, the students made scared faces when the sound of the alarm began. When the virtual airplane dropped the bomb, the students followed the animation with their gaze, expressing astonishment by opening their eyes and mouth wide (A1, (Figure 3.1)). After the activity, one child concluded: "*The people must have been scared. They probably couldn't eat or sleep.*" In the interview, one student explained that the civilians during the war constructed the shelter because they were afraid of the bomb attacks. Second, for the activity *children's drawings about the war* (A6, (Figure 3.6)), the students explained that the drawings transmitted the feelings of sadness and compassion. One child explained: "*I felt bad because the children thought a lot about the war and how they suffered.*" Third, about the activity *infirmary* (A4, (Figure 3.4)), one student expressed that he was particularly worried about the virtual person on the stretcher and imagined that this person could die during the transport. This comment may indicate that the children could empathize with the health conditions of injured civilians during bomb attacks and understood the time pressure to bring them to the hospital. Furthermore, we could observe that the WaS activities prompted collaborative interactions such as group discussions between the participants. For instance, during the 'bomb attack' activity (A1), the students discussed which

emotions the civilians experienced during bomb attacks. In the *radio* activity (A3, (Figure 3.3)), they talked about historical events during the war. The activities also encouraged group participation. For instance, during the activity *behavioural rule sign* (A5, (Figure 3.2)). One child pointed towards the marker at the wall and immediately the sign appeared. Several children who were sitting on the benches stood up and started to read out aloud the content of the sign. The guide explained that the sign was original and it was timeworn. She encouraged the children to read it out again and pointing at different words of the sign. She helped them read those that they could not decipher. Some collaborative activities also fostered children's abilities in perspective-taking. For instance, the activity *drawings of children about the war* (A6) allowed the students to compare different viewpoints on war in two different countries. In the activities *infirmary* (A4) and *construction of the shelter* (A5), the students performed similar actions to those of civilians during the war. The activity *infirmary* (A4) is also an example of the potential of the prototype to promote a 'notion of effort' [15] that may help children to empathize with sensations that civilians experienced during the war. In other words, due to the technical features of the prototype, in this activity the children had to move very slowly, carefully to carry and stabilize the virtual stretcher. Similarly, during the activity *construction of the shelter* (A5) one student in the interviews described the building method as difficult. Another child mentioned that you needed a lot of patience for the construction. At the same time, the activities supported explanations of the guide. For instance, students and a teacher pointed at different details within the projected drawings (A6, (Figure 3.6)). In the activity *construction of the shelter* (A5, (Figure 3.5)), the guide instructed the students to collaboratively construct brick by brick a virtual arch. The projection helped the guide to better explain the typical way of building in the region of Barcelona. Finally, the activities promoted spatial awareness by directing the gaze of the observer through the physical characteristics of the shelter, e.g. the trajectory of the airplane dropping the bomb on the ceiling of the shelter (A1). Another example is the *infirmary* activity (A4). While the students carried a virtual stretcher out of the shelter, their attention was directed towards one specific physical characteristics of the space. The focus was on the rounded finish around the corners on the walls that allowed a faster and safer transportation of injured people during the war.

Socio-cultural values Our analysis of students' understanding of socio-cultural values in relation to the historical context focused on the three categories human dignity, solidarity, and pluralism. Our results showed that the students were able to describe the historical context in a more nuanced way after the visit and in relation to the three socio-cultural values. For instance, the students extended their comprehension of the terms citizen association and passive defence. Furthermore, the aspects described were often related to the content of the WaS learning activities performed in the shelter (e.g. solidarity - A4, A5). Our findings are based on comparing students' worldviews before and after the visit, extracted from open-ended questionnaires, observations during the Pictionary Technique and the collage activity.

A) Students' worldviews before the visit Our results based on the open-ended questionnaires showed that the students described emotions and situations that sharply conflicted with the socio-cultural

value *human dignity* in a democratic society. They described expressions that were related to the notion of fear, sadness, nervousness, loneliness, hunger and the will to live. For instance, they referred to destroyed towns, situations of mortal danger, experiencing bomb attacks, losing relatives and friends, among others. Additionally, our observations during the Pictionary Technique revealed that the children expressed these negative feelings also on a multimodal level, i.e. we analysed students' body language while performing the activity and the facial expressions that the students used for figures in the drawings. The findings depicted that the students used facial expressions, both through mimic and in their drawings, illustrating feelings such as fear and sadness to express the emotional perspective towards concepts such as soldier, war and refugee. Finally, the group discussions indicated that even before the visit of the shelter the students could imagine and empathize with civilians during the Spanish Civil War. However, they were not aware of the long-term effects for the war survivors and its relevance for the Spanish society today. Our results based on the open-ended questionnaires showed that the students described different situations of civilians during the Spanish Civil War that were related to the concept of *solidarity*, i.e. sharing the same interests, curing injured people, supporting other's escape, sharing supplies and constructing a shelter. During the Pictionary Technique, a student who drew citizen association attributed the term to the idea of "help" between neighbours. A boy drew a bomb attack on a house and neighbours rushing to help and putting out a fire. As a representation of the term individualism, a boy drew one person wearing a cap and holding a gun. The figure confronted a group of three other people piled up as a human tower. The first person was drawn significantly larger than the other figures. This finding points towards the representation of power relations between different figures and illustrates the child's sensations of courage and cohesion when acting collaboratively in a group. On the other hand, it depicts that the child struggled to put the term into the historical and political context. The analysis of the open-ended questionnaire did not depict any aspects of children's worldviews on the socio-cultural value *pluralism*. Our results of the Pictionary Technique showed that the students correctly understood and represented the term protest. One student showed the term by drawing several people carrying banners. The term coup was related to the idea of "attack" and "war". The terms dictatorship, exile, repression and passive defence that were related to the value pluralism were rejected during the game and group discussion. These findings highlight that the students had difficulties grasping more complex concepts related to pluralism before the visit.

B) Students' worldviews after the visit Our findings of the open-ended questionnaire related to the concept *human dignity* showed that the students reported that they felt sad and compassion with the civilians because they suffered during the war. The students made high-level conclusions on concrete aspects of the war. For instance, they stressed that the city of Barcelona had been the testbed for the Italian air force. To express the meaning of the term collective trauma they used in the collage activity four photos (P4, P7, P9, P11). In the interview, the students described the same aspects represented in the photos. Overall they referred to situations that could have traumatized the civilians, e.g. (1) not being able to finish a shelter before the bomb attack begins, the need to escape or the

need to fight. These outcomes showed that the students were able to empathize with concreted situations and high level issues related to the term collective trauma. Our findings in the open-ended questionnaire showed that the students described situations of civilians during the Spanish Civil War that were related to the concept of *solidarity*, i.e. they reported that (1) bomb attacks were specific situations when civilians needed help, they highlighted that (2) injured people were supported by other civilians, (3) they described constructing a shelter as a collaborative activity between the civilians. To express the meaning of the term citizen association, the students used nine photos (P1, P2, P4, P7, P10, P11, P12, P13, P15). In the interview, the students described aspects were related to concrete actions of civilians, e.g. constructing shelters and helping injured people. They also highlighted the meaning that they associated with the term solidarity such as: teamwork, power, willingness and compassion. These outcomes illustrate that students were capable of grasping the essential qualities of the socio-cultural value solidarity. Our findings in the open-ended questionnaire related to the concept *pluralism* showed that the students reflected upon the shelter as a form of passive defence. To express the meaning of the term passive defence, the students used six photos (P2, P4, P10, P7, P12, P15). In the interview, the students mentioned aspects that described the meaning of the term, e.g. (1) defend themselves without weapons and (2) hiding in a shelter or manifestations. In addition, one child mentioned that currently, many people participated in manifestations. He was referring to the current events in Barcelona in relation to the request of independence of Catalonia. These outcomes point towards the possibility that the students made connections between historical events and present political situations in Barcelona. Summarizing, our findings showed that the learning activities based on WaS supported students' emotional experience and multiple interactions between them. Furthermore, it improved students' SEL in relation to the three socio-cultural values: Human dignity, solidarity and pluralism.

5 DISCUSSION

The present case study highlights the benefits of enhancing SEL in the context of guided visits of a Spanish Civil War shelter using the WaS paradigm. Our findings showed improvements in students' capabilities (1) to reflect upon high level-issues that conflicted sharply with the value human dignity, (2) to grasp some of the essential qualities of the socio-cultural value solidarity and (3) to connect historical events with present political situations in Spain. Based on our findings, we developed four design recommendations for educational VH experiences based on the WaS interaction paradigm to complement traditional guided visits of outdoor cultural heritage sites. (A summary of the potentials of the WaS activities can be found in the appendix (Figure 4)). We will now outline our four recommendations and point to future directions for further explorations of the WaS interaction paradigm in these contexts.

Design Recommendation 1: Supporting students' competences in communication and collaboration through balancing performance roles Our first design recommendation proposes that the WaS supports learning activities that embed the need for respect and tolerance in performance roles. In this regard, previous work [2] has highlighted the importance of supporting students

competences in communication and collaboration in order to enhance their social and emotional competences. Our findings showed that activities in which one student led the tasks and others joined as active spectators or role-play activities between several students supported group participation and group discussions. During these activities, the students needed to show respect and tolerance for each other which are considered essential skills to understand the meaning of socio-cultural values. Future work may explore further how group dynamics of different types of learning activities can support students in developing these skills. One concrete example for our prototype could be activities that require conversational turn-taking [6] and the negotiation of specific tasks.

Design Recommendation 2: Supporting students' competences in critical thinking through perspective-taking and effortful embodied explorations Our second design recommendation proposes that the WaS helps students' to think critical about historical events during guide-student interactions, perspective-taking activities and effortful embodied explorations [15]. In this regard, scholars have stressed the benefits for adult-child interaction to facilitate children's reflection upon historical events [31]. Prior research has also highlighted opportunities for learning activities focusing on the enactment of other people's actions in the past [12] and perspective-taking [12, 20] as strategies to support children's development of empathy. Our findings showed that the guide's explanations during the visit were supported by the different WaS activities. Collaborative activities, in which the students' were asked to compare different contents, helped them to take different viewpoints of people's situations and opinions. Particularly, activities that promoting the sensation of physical effort of other people's actions helped the students imagine how civilians experienced the shelter during the war. Future work could include activities that allow the students to reflect upon these experiences and opinions in a more nuanced way. One concrete example for our prototype could be a role-play activity that deals with political aspects around the Spanish Civil War, e.g. simulating the conflict between Republicans and civilians that were in favour of the dictator regime.

Design Recommendation 3: Contextualizing historical contents with spatial characteristics Our third design recommendation proposes the need for supporting students' exploration of the physical environment in meaningful ways in order to discover new layers of the educational experience. In this regard, scholars have argued for the need of contextualization of student's learning outside the classroom [4]. Our findings point towards the potential of the WaS to contextualize educational contents by promoting students' spatial awareness on specific characteristics of the shelter. Future work may explore the needs for balancing between knowledge that is situated within our experiences [3] and the possibilities to direct the gaze towards specific characteristics of a space. One concrete example for our prototype could be to include after the *bomb attack* activity another task in which the students are encouraged to explore the expansion of a shrapnel on the wall of the shelter. The activity could make the students aware about the specific way of constructing the tunnel in a zig-zag shape and where civilians needed to stand to be safe from the bomb.

Design Recommendation 4: Moral and ethical considering of digital augmentation Our fourth design recommendation proposes to consider how digital augmentations may impact students'

emotional experience and its long-term effects on children's emotional development. In this regard, previous work on cultural heritage experiences for children used the environmental and social context for augmenting the emotional engagement [19, 23]. Other scholars stressed the importance for special care and sensitivity of the context [10] and implications for the moral components using visualizations [8]. In our study, the audiovisual material used had been codesigned together the museum curators and in alignment with existing education contents for K-12 students. Our findings showed that several learning activities evoked feelings related to fear, sadness and traumatizing situations of civilians during the war. Despite our intentions to support students' empathy on civilians' experiences, we believe there is a need for a critical reflection upon the age-appropriateness of the audiovisual contents. Future work should specifically look into which cognitive impact different types of representations such as real-life photos, drawings, and animations have on children's emotional experience and develop guidelines for age-appropriate contents in the context of history education.

Limitations and conclusions This project contributes to the body of advanced interfaces by proposing four recommendation to design educational experience based on the WaS interaction paradigm in the context of cultural heritage. Our recommendations are built on the outcomes of a case study with 16 primary students in Barcelona. Together, the recommendations serve as a starting point for engaging students' in learning activities about historical contexts. In future work, we will employ, iterate and refine these design recommendations with a more advanced prototype. We encourage researchers in the CHI community to explore these design recommendations to guide the development of educational VH learning activities using the WaS paradigm. Nonetheless, some limitations are still present. Particularly, we had difficulties to assess students' previous knowledge about the socio-cultural value *pluralism* before the visit. To tackle this issues, future research needs to explore new methods to assess children's worldviews about abstract concepts and reflect upon children's cognitive capacities at different age stages.

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A APPENDIX

Table 2: Descriptions of the photos used for the collage activity

Photo number	Description
P01	Children digging up a hole for the construction of the shelter.
P02	Two children hiding under a bridge.
P03	Two children playing in front of a destroyed building.
P04	Civilians on a street escaping with their families.
P05	Destroyed buildings of the city Barcelona.
P06	A bomb attack on the city centre of Barcelona.
P07	A group of civilians digging up a tunnel for the shelter.
P08	A war fighter in the air.
P09	Soldiers at the front
P10	A woman volunteering as a nurse in the shelter.
P11	A group of people carrying an injured person on a stretcher.
P12	A group of women demonstrating for their rights.
P13	Three women sitting on the benches in the shelter
P14	A person waiting during the bomb attack in the shelter.
P15	Empty tunnels of the shelter.

Activity	Content Type			Interactions initiated through WaS							Potentials for the educational experience						
	Still image	Animation	Audio	Performance	Performance one student	Performance in pairs	Role-play in pairs	Role-play in group	Effortful actions	Directed gaze in physical space	Emotional impact	Awareness Space	Group participation	Group discussions	Supported explanations	Perspective-taking	Notion of effort
A1: Bomb Attack		X	X	X						X	X	X		X	X		
A2: Behavioural rule sign	X				X								X		X		
A3: Radio documentary	X		X		X									X	X		
A4: Infirmary	X			X				X	X	X	X	X	X		X	X	X
A5: Construction shelter	X			X			X		X	X		X			X	X	X
A6: Children's drawings	X					X					X				X	X	

Figure 4: Overview potentials of the WaS activities